

# Servlet Program installation

## 1.How to check Servlet JAR file in Your System is Present or not ?

```
ls /usr/share/tomcat/lib/
```

## 2. Setup Bashrc file

```
vim ~/.bashrc
```

```
export CLASSPATH=$CLASSPATH:../usr/share/tomcat/lib/tomcat-servlet-4.0-api.jar
```

```
source ~/.bashrc
```

## 3.Where is Servlet Program is Present

**/var/lib/tomcat/webapps/myapp/WEB-INF/classes/  
.java file (Servlet)**

```
|— index.html          ← your HTML file
|— WEB-INF/
|   |— classes/
|       |— WelcomeServlet.class ← compiled servlet
```

## 4. Where is html or xml is saved

**/var/lib/tomcat/webapps/myapp**

Open Terminal Here.

## Type 1: Servlet with HTML

**Vi index.html**

```
<!DOCTYPE html>
<html>
<head>
  <title>HTML to Servlet Example</title>
</head>
<body>
  <h2>Enter Your Name</h2>

  <form action="welcome" method="POST">
    <label for="username">Username:</label>
```

```
        <input type="text" id="username" name="username">
        <br>
        <input type="submit" value="Submit">
    </form>

</body>
</html>
```

## 5. Where is Servlet Program is write

**/var/lib/tomcat/webapps/myapp/WEB-INF/classes**

**Open Terminal Here.**

### **Vi WelcomeServlet.java**

```
import java.io.IOException;
import java.io.PrintWriter;

import javax.servlet.ServletException;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;

@WebServlet("/welcome")
public class WelcomeServlet extends HttpServlet {

    protected void doPost(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {

        response.setContentType("text/html");

        PrintWriter out = response.getWriter();
        String userName = request.getParameter("username");

        out.println("<!DOCTYPE html>");
        out.println("<html>");
        out.println("<head><title>Welcome Page</title></head>");
        out.println("<body>");
        out.println("<h2>Hello, " + userName + "!</h2>");
        out.println("</body>");
        out.println("</html>");
    }
}
```

```
        out.close();
    }
}
```

**Compile Servlet :**

1. `javac WelcomeServlet.java`
2. `sudo systemctl restart tomcat`

**"Note: Start the Tomcat server each time before running a servlet program."**

**6.Output**

**Go on Browser**

<http://localhost:8080/webapp/index.html>

<http://localhost:8080/webapp/welcome>

## **Type2: SERVLET with XML**

**Imp Note : Every time you have to give class name of your servlet in Web.xml**

**1.How to write servlet program with .xml file**

**Location**

**Tomcat/**

```
├── webapps/
│   ├── myapp/
│   │   ├── WEB-INF/
│   │   │   ├── web.xml
│   │   │   ├── classes/
│   │   │   └── info.class(info.java file)
```

**Go To,**  
**`/var/lib/tomcat/webapps/myapp/WEB-INF/`**

**Open Terminal Here.**

## **Web.xml**

```
<?xml version="1.0" encoding="ISO-8859-1"?>
<web-app>
<servlet>
<servlet-name>InfoServlet</servlet-name>
<servlet-class>InfoServlet</servlet-class>
</servlet>
<servlet-mapping>
<servlet-name>InfoServlet</servlet-name>
<url-pattern>/InfoServlet</url-pattern>
</servlet-mapping>
</web-app>
```

**Go To,**

**/var/lib/tomcat/webapps/myapp/WEB-INF/classes/  
.java file (Servlet)**

## **InfoServlet.java**

```
import java.io.*;
import javax.servlet.*;
import javax.servlet.http.*;
import java.util.*;

public class InfoServlet extends HttpServlet {

    public void doGet(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {

        response.setContentType("text/html");
        PrintWriter out = response.getWriter();

        out.println("<html><body>");
        out.println("<h2>Client Information</h2>");
```

```

// Client Information
out.println("IP Address: " + request.getRemoteAddr() + "<br>");
out.println("Browser Type: " + request.getHeader("User-Agent") + "<br>");

out.println("<h2>Server Information</h2>");

// Server Information
out.println("Operating System: " + System.getProperty("os.name") + "<br>");
out.println("Server Name: " + request.getServerName() + "<br>");

// Loaded Servlets
ServletContext context = getServletContext();
Set<String> servlets = context.getServletRegistrations().keySet();

out.println("<h3>Loaded Servlets:</h3>");
for (String servletName : servlets) {
    out.println(servletName + "<br>");
}

out.println("</body></html>");
out.close(); // Good practice
}
}

```

( **Note** :When writing a Servlet program, make sure that the **class name in your Java file** exactly matches the **class name you mention in web.xml**. Also, ensure that the **URL pattern in web.xml** is correctly set, because this is the address you will use in the browser to access your servlet.)

**1.Compile it(java file) and run restart tomcat Server**

**2.Javac info.java**

**3.sudo systemctl restart tomcat**

**Go ON browser**

**How to run on browser**

<http://localhost:8080/myapp/info>